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REVIEW

Behavior Science Contributions to Public Policy: an Introduction to the Special Section

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Abstract

Behavior science has a long history of influencing public policy. Numerous scholars have used behavioral principles in experimental and applied research to examine the potential impact of local, state, and federal policies across socially important problems and goals. The utility of behavior science in public policy continues to flourish, and translational behavioral research will remain a critical component of effective policy development and implementation. The articles in this special section highlight diverse examples of applied research in various areas, such as intellectual disabilities, substance use, and greenhouse gas emissions. In addition, this special section includes findings from experimental research demonstrating the benefits of using demand curve analysis and behavioral procedures such as *nudging* and *boosting* to facilitate effective policy change. Together, these articles offer diverse exemplars of behavior science's importance in public policy development and implementation.

Keywords Public policy · Behavior science · Intellectual disabilities · Substance use · Environmental health · Demand curve analysis

Behavior science has long provided empirical findings that inform public policy. Fawcett et al. (1988) provided illustrative examples of how behavior scientists can become more involved in policy-making. Since then, numerous experimental and applied studies have demonstrated the utility of behavioral principles and technologies in improving socially important outcomes. In addition to informing policy with

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widespread social implications (Fischhoff, 2021; Hursh & Roma, 2013), behavior science offers tools and methodologies for estimating the impact of existing and future policies (Chetty, 2015). Furthermore, scholars have identified processes for engaging in policy-making in ways that are grounded in behavior science (Mattaini et al., 2020; McConnell, 2021; Todorov & Freitas Lemos, 2020). As behavior science continues to expand its areas of application, translational behavioral research in policy implications will remain a critical and fruitful endeavor for the field. As such, the purpose of this special section is to provide illustrative examples of experimental and applied research that integrate behavior science and policy. The works published in this special section span the domains of intellectual disabilities, substance use, experimental analysis, and environmental science. Here, we briefly introduce the articles in the section.

Houck and Dracobly argue that policies have been developed to address multiple historically marginalized communities, including Indigenous people, military veterans, women, and children. However, many existing legislative policies in the United States fail to address the specific and unique needs of individuals with intellectual disabilities, particularly those with intersecting identities and individuals with trauma histories. To address this issue, Houck and Dracobly provide several policy recommendations to better support people with intellectual disabilities, such as: (1) acknowledging trauma and sharing information; (2) requiring individualized assessment and services across the lifespan; and (3) funding research on the assessment and treatment of stress disorders among people with intellectual disabilities. Such recommendations can be readily integrated into behavior analytic practices through continuing education credits related to trauma and behavioral health and using validated measures to identify potentially traumatic events. A more long-term policy change includes requiring insurance coverage to address trauma across the lifespan within the practice of applied behavior analysis for persons with intellectual disabilities.

A similar application of behavior science in policy is in substance use. DeFulio suggests that leveraging smartphone technology to deliver contingency management interventions has great potential to reduce substance use disparities. At present, for many substance use interventions, costs and referrals remain a substantial barrier. For example, obtaining referrals for substance use treatment are often hindered by long wait times, lack of provider or facility availability, and costs of such treatment. As DeFulio notes, smartphone-based contingency management interventions may be advantageous given that approximately 85% of U.S. adults own smartphones (Pew Research Center, 2021). In addition, the personalized nature of smartphone-based technologies can mitigate potential stigmatizing situations (e.g., being seen entering or exiting a clinic or facility). As technology continues to be integrated in health, policies that support technology-based behavior science facilitate increased collaborations among behavior science, medicine, and public health to reduce health inequities.

It must be noted, however, that policy development, adoption, and implementation do not come without critical considerations of resources such as time. For instance, long wait times to receive medical, dental, or social services can reduce the value of seeking such services. Schwartz and Hursh conducted a series of

experiments that demonstrate the utility of using demand curve analysis to examine time-restricted behavior. In addition, they demonstrated that including time perception parameters can aid in understanding relationships between time delays and consumption as part of an overall evaluation of demand impacts. Schwartz and Hursh identify several implications for policy decisions in areas such as accessing financial aid for higher education, testing for COVID-19 and HIV/STIs, and accessing and using public versus private transportation. Another area in which the experimental findings hold substantial application is in accessing food assistance programs, particularly given the long wait times present in completing applications and obtaining services.

Tagliabue offers greater insight into how behavioral operations such as *nudging* and *boosting* can support policy implementation. As Tagliabue describes, nudging provides policy makers with readily implementable opportunities such as reminders, leveraging social norms, or disseminating warnings. Boosts not only provide positive reinforcement for choice behavior, but they can also work in tandem with nudges to inform and boost policies aimed at reducing health disparities.

Bonner et al. then offer a critical analysis of the need for additional funding for behavior science approaches to reduce greenhouse gas emissions. Although funding for environmental health has increased substantially, the Government Accountability Office has not funded any explicit behavior science research. Furthermore, a systematic investigation conducted by the authors found that few projects funded by five federal agencies (National Science Foundation, Environmental Protection Agency, Department of Energy, National Institutes of Health, Centers for Disease Control and Prevention) were explicitly grounded in behavior science. Furthermore, an even lower percentage of those funded studies involved experimental manipulations. From a policy perspective, funding agencies can begin prioritizing and allocating grant funding in areas such as assessing the impact on emissions, policy adoption, and policy implementation; integrating behavioral technologies in examining the impact of greenhouse gas emissions; and developing and testing comprehensive community interventions. Behavioral scientists can also support these policy recommendations by advocating more strongly for supporting experimental and applied behavior science in addressing greenhouse gas emissions.

The articles in this special section offer clear applications of how behavior science can inform public policy and specific recommendations for future integration. We hope this special section promotes further experimental and applied research in using behavior science to develop, implement, and evaluate policies that have implications for community and population health. As public interest in health equity continues to grow, behavior science can provide empirically driven policy recommendations that not only promote positive behavior change, but also improves health conditions across multiple ecological levels.

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Declarations

Conflict of Interest We have no known conflict of interest to disclose.

Data Availability Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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